THE CANONICAL SHEAF RESTRICTS TO DEGREE ZERO ON A RATIONAL BRIDGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument:

Editorial note. The label lemma-rational-bridge-zero names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The rational bridge setup. Same chapter, Example \label{example-rational-bridge}. Proof lines 6321-6323 and 6327-6330: X' is the closed subscheme as in the example and k"/k'/k are the field extensions of the example; deg(omega_C) = -2[k' : k] because C has genus 0, and deg(C \cap X') = [k" : k] = 2[k' : k].
- Canonical sheaf on closed subschemes of nodal curves. Same chapter, Lemmas \label{lemma-closed-subscheme-reduced-gorenstein}, \label{lemma-facts-about-nodal-curves}, \label{lemma-closed-subscheme-nodal-curve}. Proof lines 6323-6326: the short exact sequence 0 -> omega_C -> omega_X|_C -> O_{C \cap X'} -> 0.
- Riemann-Roch for curves. Same chapter, Lemma \label{lemma-rr}. Proof lines 6327-6330: C has genus 0, so deg(omega_C) = -2[k' : k].

Source reference. Original reference 0E3M: sources/stacks-curves/pinned-original-source.tex, source lines 6201–6201. Prerequisite pointer invoked at proof line(s) 6321; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E38: sources/stacks-curves/pinned-original-source.tex, source lines 4794–4794. Prerequisite pointer invoked at proof line(s) 6332; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E34: sources/stacks-curves/pinned-original-source.tex, source lines 1012–1012. Prerequisite pointer invoked at proof line(s) 6330; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0E37: sources/stacks-curves/pinned-original-source.tex, source lines 4763–4763. Prerequisite pointer invoked at proof line(s) 6331; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS6: sources/stacks-curves/pinned-original-source.tex, source lines 1218–1218. Prerequisite pointer invoked at proof line(s) 6335; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Assume the singularities of X are at-worst-nodal. Let $C \subset X$ be a rational bridge (Example 0E3M). Then* $\deg(\omega_X|_C) = 0$.

Proof. Let $X' \subset X$ be as in the example. Then we have a short exact sequence

$$0 \to \omega_C \to \omega_X|_C \to \mathcal{O}_{C \cap X'} \to 0$$

See Lemmas 0E34, 0E37, and 0E38. With $k''/k'/k$ as in the example we see that $\deg(\omega_C) = -2[k' : k]$ as C has genus 0 (Lemma 0BS6) and $\deg(C \cap X') = [k'' : k] = 2[k' : k]$. Hence $\deg(\omega_X|_C) = 0$. □